



Record your observations in Table 13.2.

Table 13.2. Effect of cations on flocculation of a clay suspension.

Added cation	Relative Size & Settling Rates of Floccules
K ⁺	
Na ⁺	
Ca ²⁺	
Al ³⁺	
Check	

Activity 4. Determining CEC by replacing adsorbed cations.

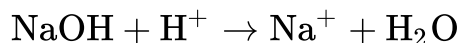
In this activity, you will titrate the filtrate with a 0.01 molar solution of NaOH using phenolphthalein as an indicator. Phenolphthalein changes from colorless to faint pink when the quantity of OH⁻ ions added via the NaOH equals the quantity of H⁺ ions in the solution (that is, when the pH is raised to 7). For this activity, assume the soil samples have been extracted and the filtrates are now available for analysis.

1. Place 10 ml of each filtrate into separate 125 ml flasks. This 10 ml quantity is the amount of filtrate from 1.0 gram of soil.
2. Add 10 drops of the phenolphthalein indicator.
3. Titrate the extract with the NaOH solution to a faint pink endpoint. The titration must be done very carefully to obtain meaningful results. If you put too much NaOH in the flask and get a bright pink color, discard the solution and repeat the process. In the table below, record the milliliters of NaOH solution used to achieve the endpoint.



Calculate the CEC and record your data in Table 13.3.

Here is an example of how to calculate the CEC, assuming 2.5 mL of NaOH was required to achieve an end point. The reaction occurring during titration is



Thus, one mole of NaOH reacts with one mole of H⁺. Therefore, at the phenolphthalein end point, moles of NaOH added = moles of H⁺ in solution.

The solution of 0.01 molar NaOH contains 1 cmol charge per liter (1 cmol_c/L). Therefore 2.5 mL NaOH contains

$$\text{cmol}_c \text{ of NaOH} = 2.5 \text{ mL NaOH} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{0.01 \text{ mol NaOH}}{1 \text{ L}} \times \frac{1 \text{ mol}_c}{1 \text{ mol NaOH}} \times \frac{100 \text{ cmol}_c}{1 \text{ mol}_c} = 0.0025 \text{ mol}_c \text{ NaOH}$$

Thus, the CEC is

$$\frac{\text{cmol}_c}{\text{kg soil}} = \frac{0.0025 \text{ cmol}_c}{1 \text{ g soil}} \times \frac{1000 \text{ g soil}}{1 \text{ kg soil}} = \frac{2.5 \text{ cmol}_c}{\text{kg soil}}$$